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DriveValue AI

Explainable Machine Learning Framework for Predicting Used Vehicle Prices

ZAIO Data Science Capstone Project | 2026

DriveValue AI — End-to-End Solution & Methodology

Project Overview

DriveValue AI estimates used vehicle asking prices using historical listing data, following the CRISP-DM methodology from business understanding through deployment. The framework combines Random Forest Regression for predictive accuracy, SHAP Explainable AI for model transparency, and a Streamlit web application for interactive deployment. Together, these components deliver transparent, consistent, and data-driven vehicle price predictions that support objective decision-making in the used vehicle marketplace.

Business
Problem

The Business Problem

Used vehicle pricing remains subjective and inconsistent, with similar vehicles commanding widely varying prices across sellers and marketplaces. This inconsistency leads to reduced profitability, slow inventory turnover, customer dissatisfaction, and unreliable decision-making for dealerships and individual sellers alike. DriveValue AI addresses this challenge by delivering an objective, data-driven pricing solution that leverages historical listing data to bring transparency and consistency to used vehicle valuation.

Project Goals

Business Objectives

DriveValue AI aims to deliver accurate, transparent, and deployable vehicle price predictions through machine learning and explainable AI.

01

Predict & Compare

Predict asking prices using ML and compare multiple regression algorithms.

02

Select Best Model

Identify and select the best-performing predictive model from evaluation.

03

Explainable AI

Improve transparency via SHAP Explainable AI for stakeholder trust.

04

Deploy & Demonstrate

Demonstrate deployment readiness with a Streamlit web application.

Used Cars Dataset

	Brand	model	Year	Age	kmDriven	Transmission	Owner	FuelType	PostedDate	AdditionInfo	AskPrice
0	NaN	NaN	2001.0	23.0	98,000 km	NaN	NaN	Unknown	Nov-24	Honda City v teck in mint condition, valid gen...	₹ 1,95,000
1	Toyota	NaN	NaN	15.0	190000.0 km	Manual	NaN	Diesel	Jul-24	Toyota Innova 2.5 G (Diesel) 7 Seater, 2009, D...	₹ 3,75,000
2	Volkswagen	NaN	2010.0	NaN	NaN	Manual	first	NaN	Nov-24	Volkswagen Vento 2010-2013 Diesel Breeze, 2010...	₹ 1,84,999
3	NaN	Swift	2017.0	7.0	NaN	Manual	second	Unknown	NaN	Maruti Suzuki Swift 2017 Diesel Good Condition	₹ 5,65,000
4	Maruti Suzuki	NaN	2019.0	5.0	NaN	Automatic	NaN	Unknown	Nov-24	Maruti Suzuki Baleno Alpha CVT, 2019, Petrol	₹ 6,85,000

```
X_train, X_test, y_train, y_test
    X,
    y,
    test_size=0.20,
    random_state=42
)

print("Training:", X_train.shape)
print("Testing :", X_test.shape)
```

```
Training: (11990, 466)
Testing : (2998, 466)
```

14,988 vehicle listings with 466 engineered features after cleaning and preprocessing. Target variable: AskPrice. Key predictors include Vehicle Brand, Model, Year, Vehicle Age, Kilometres Driven, Fuel Type, Transmission, Previous Owner, and Posted Date.

Our Structured Approach

CRISP-DM Methodology

We follow the CRISP-DM framework to ensure a rigorous, repeatable process from business understanding through deployment. Each phase builds on the previous, delivering transparent and reliable price predictions.

01

Business & Data Understanding

Define objectives, explore data structure, and assess data quality.

02

Data Preparation & EDA

Clean data, handle missing values, and visualise key patterns.

03

Feature Engineering & ML Modelling

Engineer 466 features and train regression algorithms.

04

Evaluation, SHAP & Deployment

Evaluate models, explain with SHAP, deploy via Streamlit.

Data Preparation

DriveValue AI transforms raw used vehicle listings into a structured, machine-learning-ready dataset through systematic preprocessing, ensuring reliable inputs for accurate price prediction.

01

Duplicate removal and missing value handling

Identified and removed duplicate listings; handled missing values through imputation and exclusion to ensure data quality.

02

Feature engineering with one-hot encoding

Engineered 466 features from raw data, including one-hot encoding of all categorical variables for machine learning compatibility.

03

Train-test split and feature alignment

Split data into training and test sets; aligned features to ensure consistent input structure for model deployment.

04

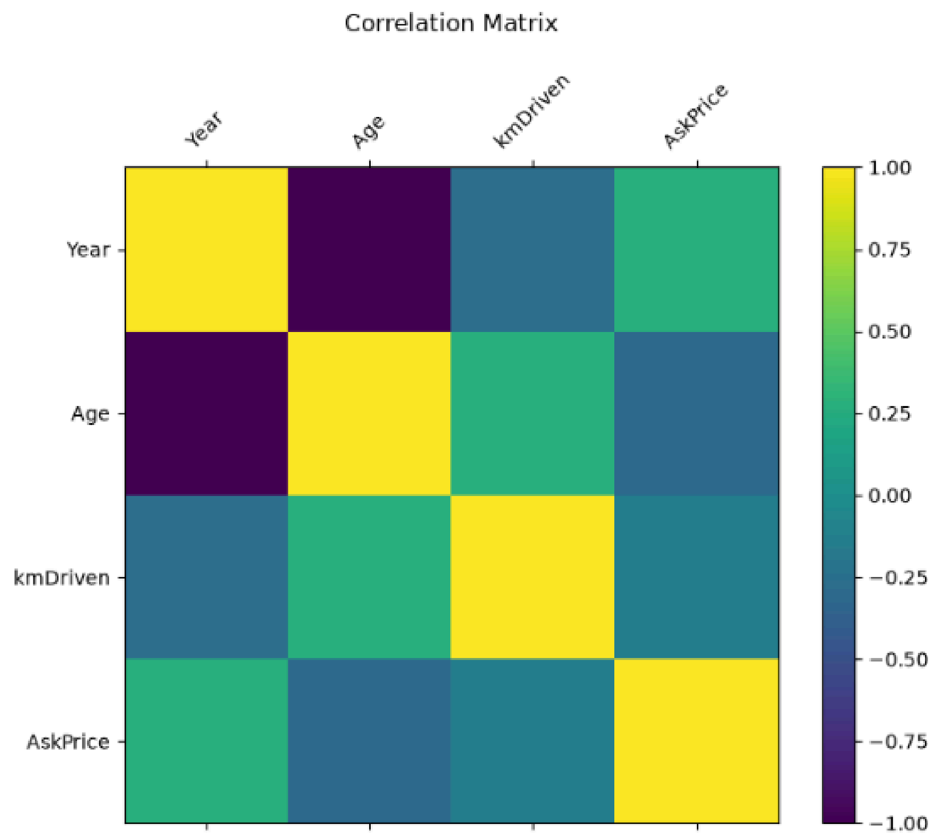
Final dataset: 14,988 observations, 466 ML-ready features

Clean dataset ready for modelling: 14,988 vehicle observations with 466 machine-learning-ready features.

Exploratory Data Analysis

Key Insights from Data Exploration

EDA revealed critical patterns in used vehicle pricing. Vehicle age, brand prestige, and mileage emerged as primary price drivers, while fuel type and transmission introduced meaningful variation. These insights directly informed feature engineering and model selection for DriveValue AI.



01

Vehicle Age Impact

Older vehicles show consistently lower asking prices across all segments.

02

Brand Premium Effect

Luxury and premium brands command significantly higher market prices.

03

Mileage Correlation

Lower kilometres driven strongly correlate with higher vehicle valuations.

04

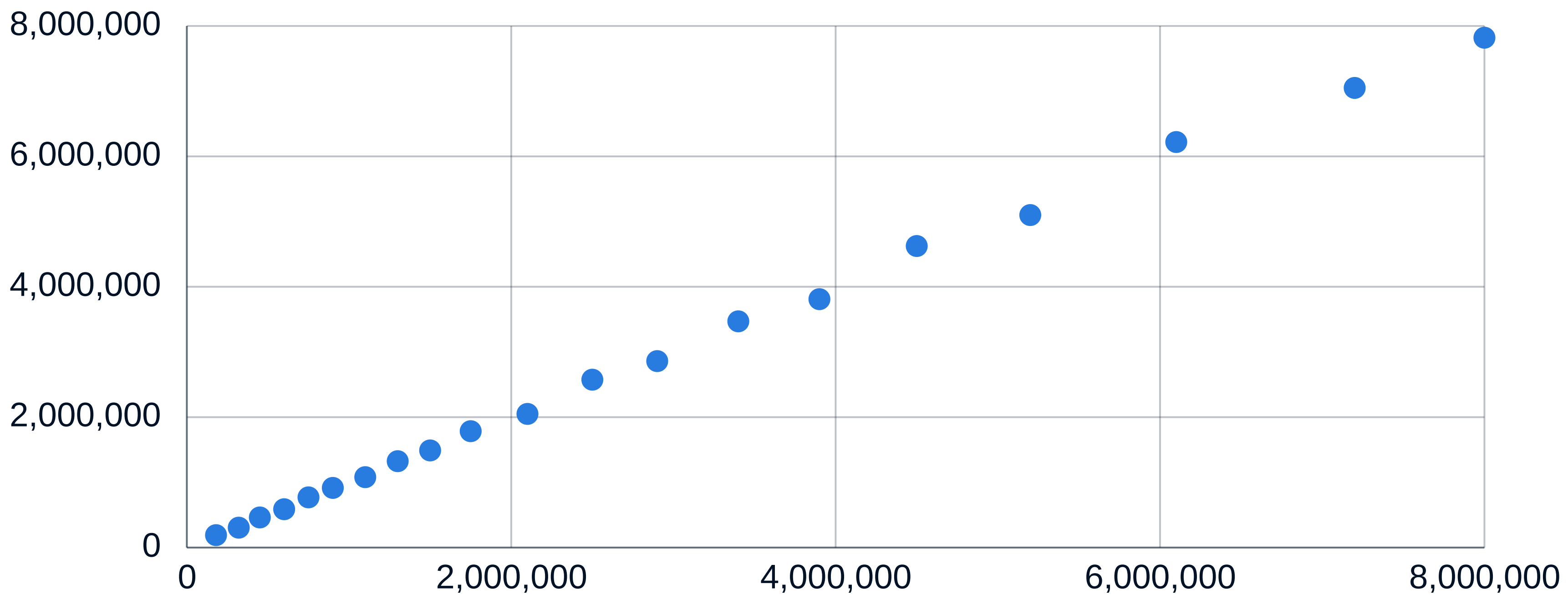
Feature Variation

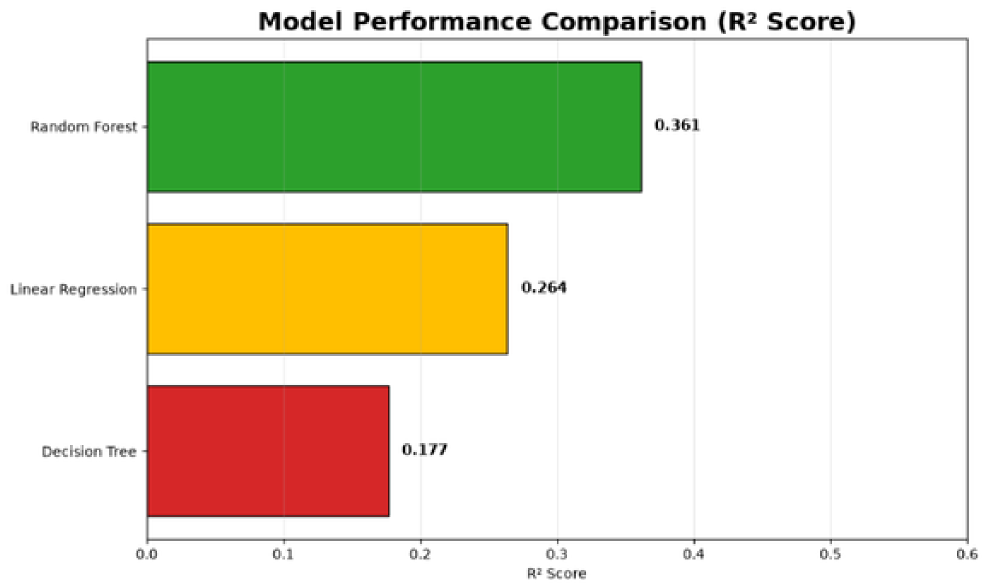
Fuel type and transmission create distinct price distribution patterns.

Algorithm Comparison & Final Model Selection

Algorithm	Strengths	Limitations
Linear Regression	Simple, interpretable baseline; fast training	**Assumes linear relationships; underfits complex patterns**
Decision Tree	Captures non-linear interactions; easy to visualize	**Prone to overfitting; unstable with small data changes**
Random Forest	Ensemble robustness; handles mixed feature types well	**Higher compute cost; less interpretable than single trees**

Random Forest Regressor Results





The Random Forest Regressor achieved the best predictive performance among the evaluated models, explaining approximately 44.6% of the variation in vehicle asking prices while demonstrating good generalisation to unseen data.

Model
Transparency &
Trust

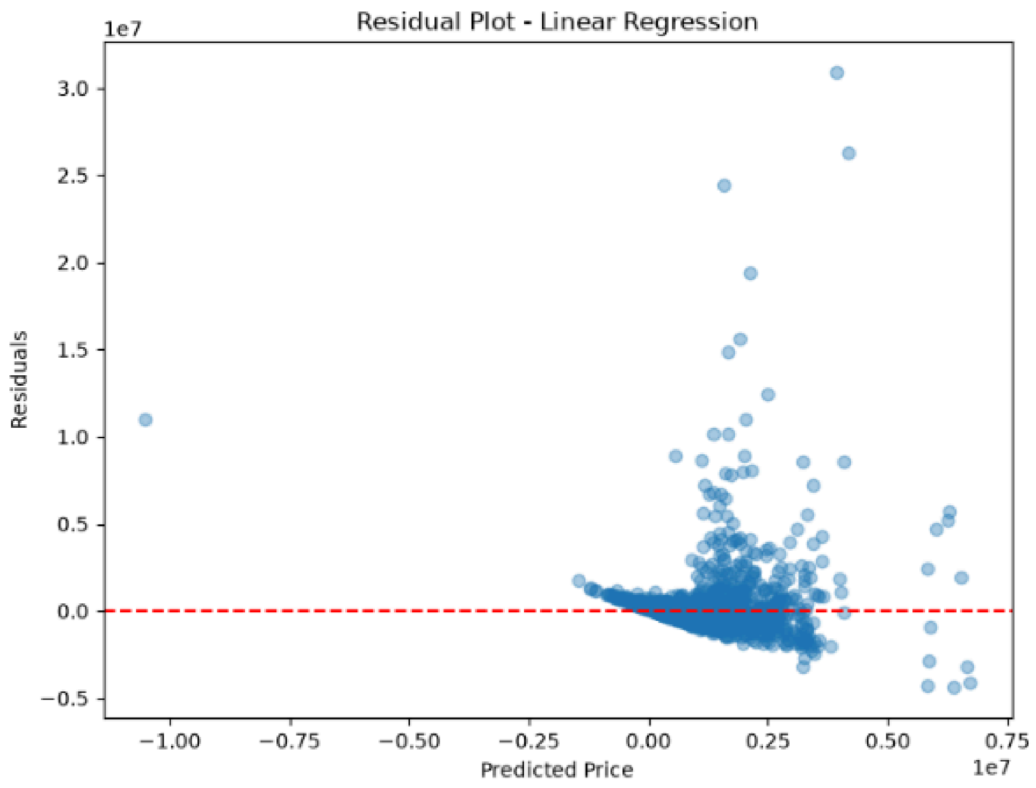
Explainable AI

DriveValue AI leverages SHAP (SHapley Additive exPlanations) to deliver full transparency behind every price prediction. SHAP explains how each vehicle characteristic influences the predicted asking price—both for individual listings and across the entire dataset.

This builds stakeholder trust by demystifying the model's decisions, enabling dealers and marketplaces to justify prices with data-driven reasoning. The most influential factors identified include Vehicle Age, Brand, Model, Year, Kilometres Driven, Fuel Type, and Transmission—aligning with domain expertise and ensuring transparent, defensible pricing.

DriveValue AI

Model Evaluation



Actual vs Predicted Price plot shows predictions closely follow the ideal line, indicating strong model reliability. Residual plot confirms errors are randomly distributed around zero with minimal bias. The model performs consistently across a wide vehicle price range, with strongest accuracy for mainstream vehicles.

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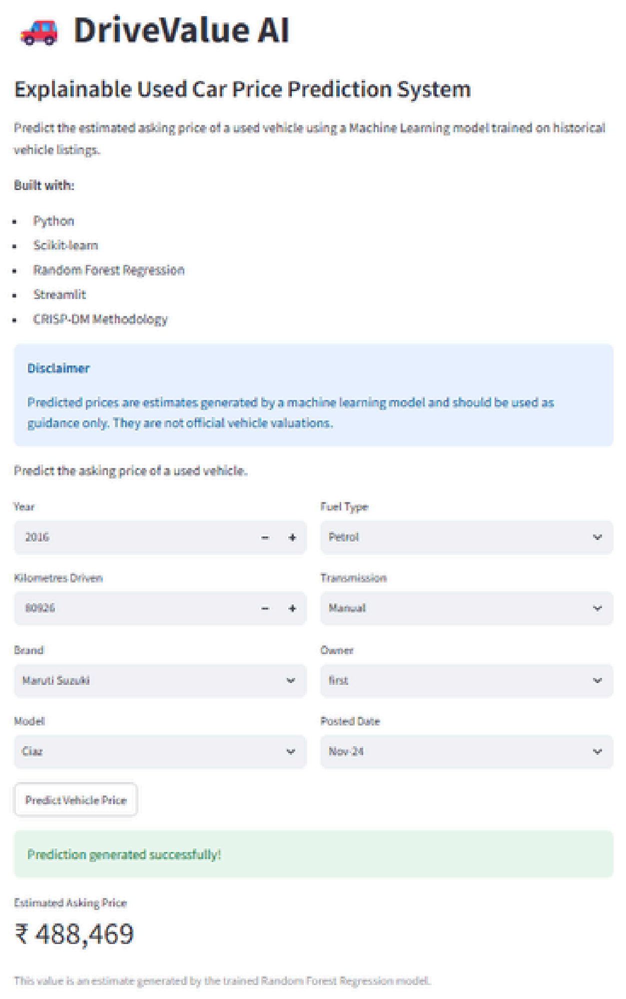
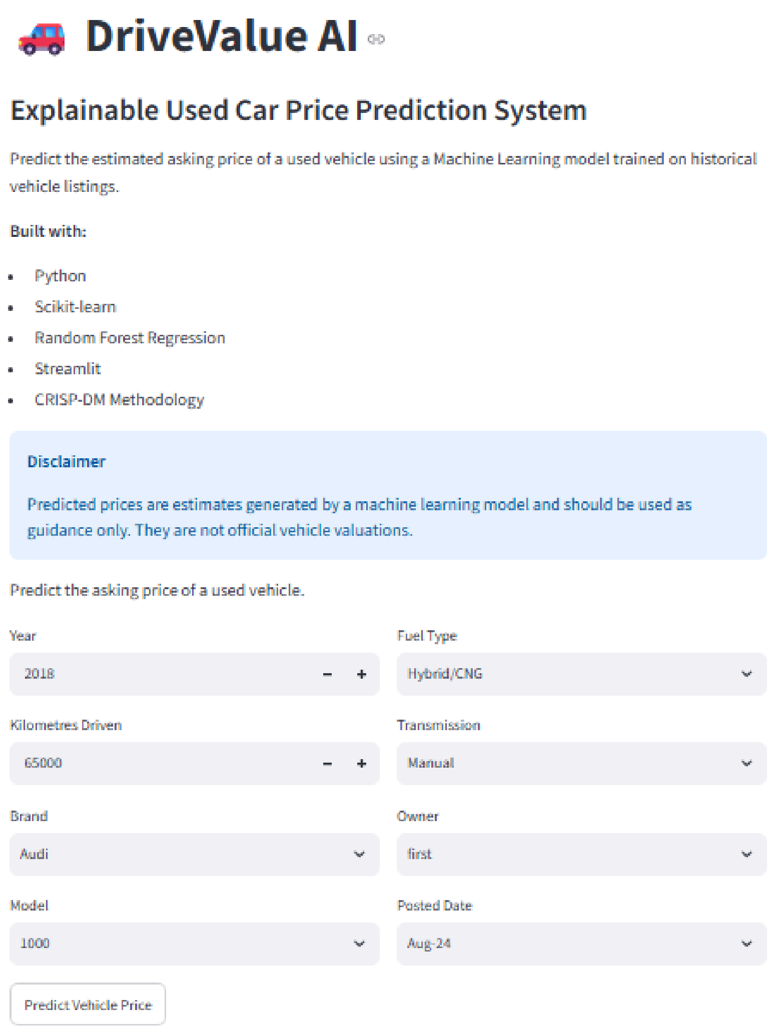
Streamlit Deployment





The DriveValue AI solution is deployed as an interactive Streamlit web application that transforms raw vehicle inputs into instant price predictions.

Workflow: Vehicle Details → Feature Engineering → Random Forest Prediction → Estimated Price.

Users enter vehicle characteristics through an intuitive interface; the app applies the same preprocessing and feature engineering pipeline used in model training, then generates a transparent, explainable price estimate in real time.
This deployment demonstrates production readiness and makes data-driven pricing accessible to dealerships and marketplace users.

Business Value

Project Impact and Benefits

DriveValue AI delivers measurable value to dealerships and vehicle marketplaces through fair, transparent, and efficient pricing powered by machine learning.

01

Fair Pricing

Enables consistent, objective vehicle pricing across all inventory.

02

Reduced Bias

Data-driven decisions minimize subjective judgment and pricing inconsistency.

03

Transparency

SHAP explainability builds stakeholder trust with clear price factors.

04

Efficiency & Confidence

Speeds dealership workflows and improves customer confidence.

DriveValue AI

Future Improvements

Planned enhancements to the used vehicle pricing solution. Scaling deployment, integrating live data, automating retraining, and advancing model performance with ensemble methods.

01

Cloud Deployment

Deploy to Streamlit Community Cloud, Microsoft Azure, or AWS for scalable, accessible hosting.

02

Live Data Integration

Integrate live vehicle marketplace feeds for real-time, continuously updated pricing predictions.

03

Automated Retraining

Schedule periodic model retraining pipelines to maintain accuracy as market conditions evolve.

04

Advanced Models & Analytics

Improve accuracy with Gradient Boosting or XGBoost, plus an interactive dealership analytics dashboard.

Project Outcomes & Achievements

Conclusion

DriveValue AI successfully demonstrated the complete CRISP-DM lifecycle by developing, evaluating and deploying an explainable machine learning solution for used vehicle price prediction. The Random Forest Regressor achieved the strongest predictive performance ($R^2 = 0.446$), while SHAP improved model transparency and Streamlit demonstrated deployment readiness for real-world use.

Questions?

Thank You

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https://github.com/JacquaM/DriveValue-AI
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